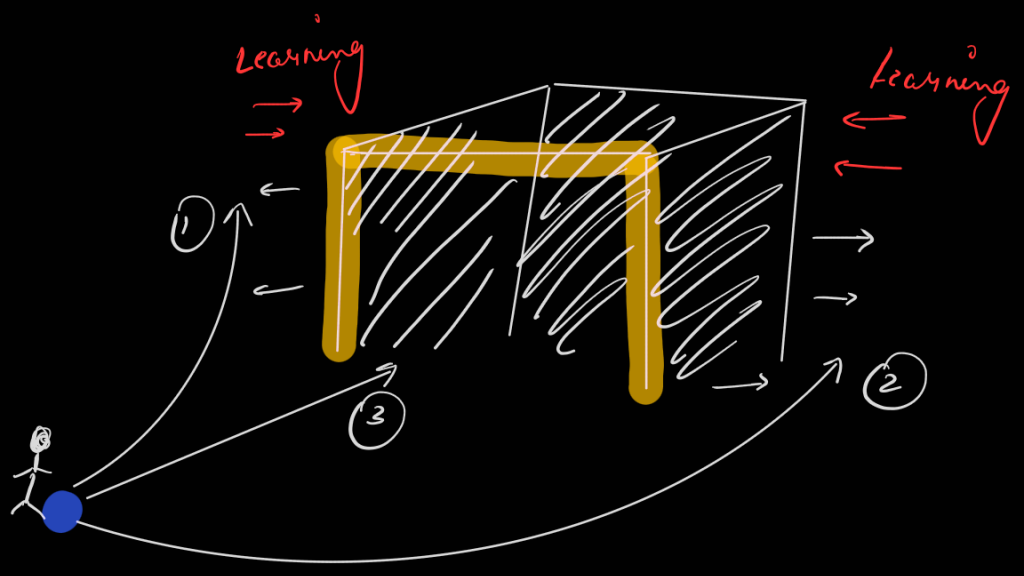
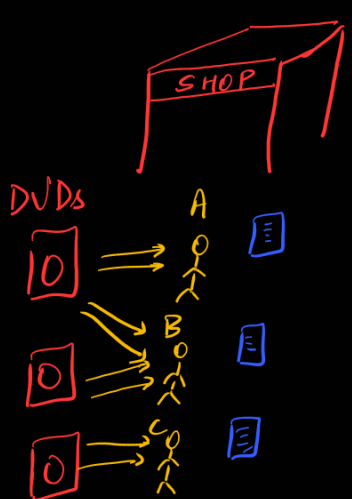
ML is EVERYWHERE!!!



ML → "Finding Patterns"

Examples → Alexa, Netflix, Zomato / Swiggy, Google Maps

Netflix → TV (Biggest)



Customer's Preferences

→ Preferences?

→ Basic info → Loc, Age, email

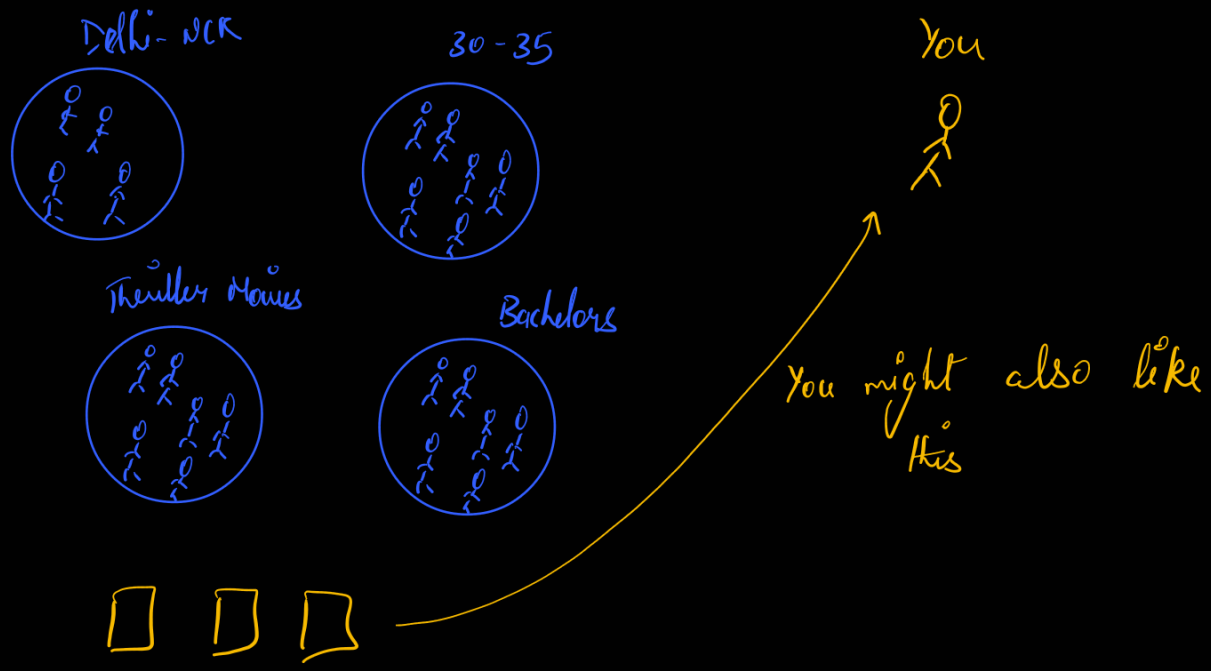
Platform Analytics

→ Weekday / Weekends

→ Morning / Nights

→ One go Vs 1-1 episodes

→ Leaving in b/w



Recommender Systems (Rec-Sys)

Amazon → Mobile → Recommendations → Screen guard.
Recommendations → Similar Phones
Recommendations → Back Cover

Swiggy | Zomato | Myntra | Gaana | Spotify.

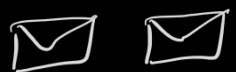
Alexa → command → understands → Perform action.

Voice → Loud
Voice → Should
Voice → Pitch
Voice → Accent
Voice → Male | Fem
Voice → Adult | Kid | Old.

Example: Spam Email → Suspi → Spam (S)
Email → Inbox Not Spam (NS)

Traditional Programming → Fails

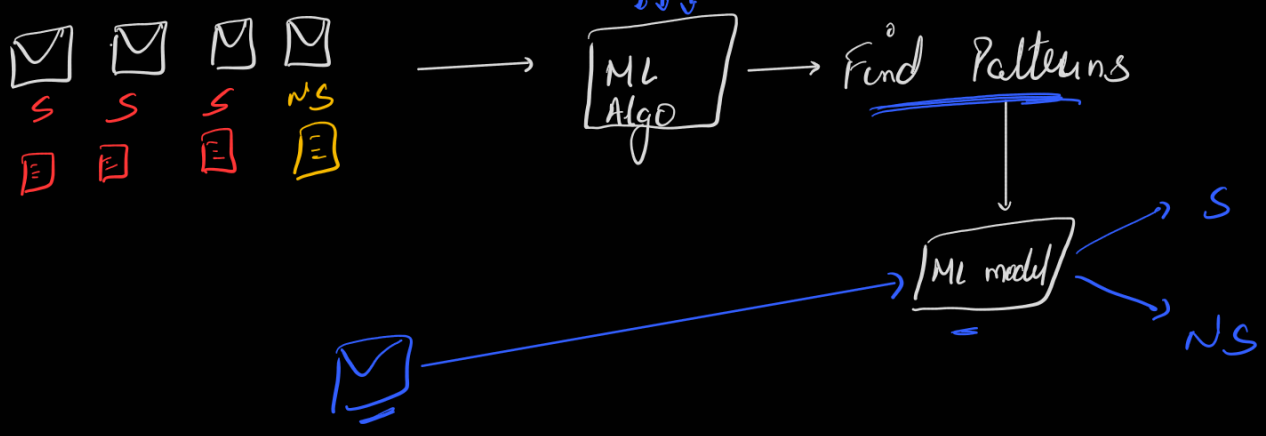
→ Keywords → "Jackpot"
"Lottery"
"Urgent"
⋮

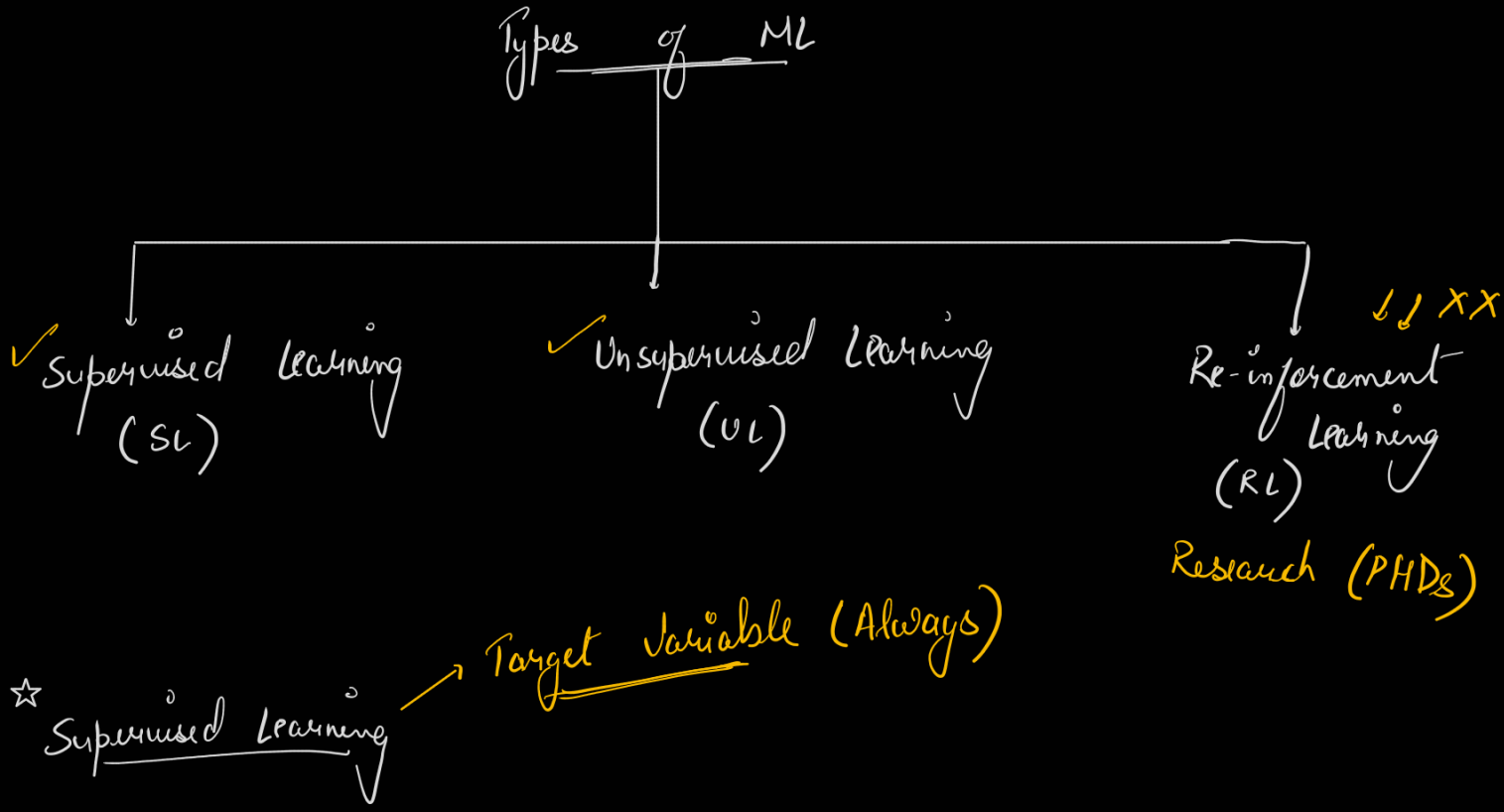

→  → No. of Bad keywords → Content 100 words → 50% → Spam

"if else conditions" → Rules

ML → Works and is scalable.

1000 → S / NS





☆ Supervised Learning → Target Variable (Always)

Problem: Can help MB to predict the house prices?

Data → ML Algo → Find Patterns → ML Model → Predictions

Indepnt Features / Features — Target Var Depn

Area (sqft)	Age	Locality	# Bathrooms	Price
1200	5	A	2	80L
900	2	B	1	20L
1200	5	B	2	60L
2400	10	A	3	1.2Cr
1500	7	B	2	?

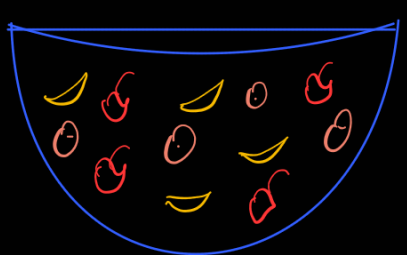
x_i Datapoint — Label y_i

"Regression" → LR, DT, RF

Unsupervised Learning → No Target Variable

"Clustering" → KMeans, DT/RF, HC

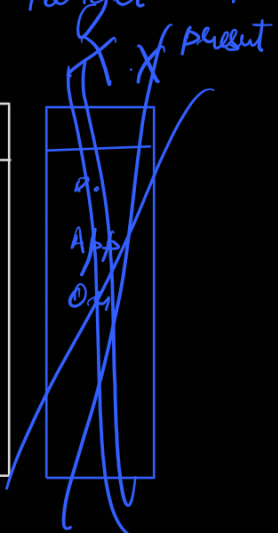
Color	Shape	Size
Y	Boat	L
Red Orange	Oval Round	M S
Y	B	L
Red Org	Oval Round	M S
Y	B	L





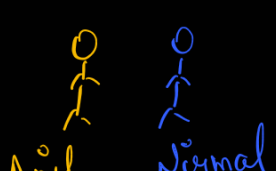






Examples → Customer Segmentation, Product Segmentation

Re-Inforcement Learning



Total points = 100
Penalty = -10
Reward = +10





Agent

Entry

Supervised Learning

Target Variable Required

- **Regression** → Predicting House Prices, Car Prices
  - Marks → Salary
- **Classification**
  - Spam / Not Spam
  - Dog / Cat
  - Emp Attrition
  - Cust Churn
  - Sedan / SUV / Hatchback
- **Forecasting**
  - Weather Forecast
  - Sales Forecast
  - Demand Forecast
  - Time Aspect

Unsupervised Learning

No Target

→ Clustering → Fruit Example

(1)

Features			Target
Color	Shape	Size	Fruit
Y	Boat	L	Banana
Red	Oval	M	Apple
Orange	Round	S	Orange
Y	B	L	B
Red	Oval	M	A
Or	Round	S	O
Y	B	L	B

→ Classification

Y B L ? Banana

Clustering

Color	Shape	Size
Y	Boat	L
Red	Oval	M
Orange	Round	S
Y	B	L
Red	Oval	M
O4	Round	S
Y	B	L

Y B L ?



ML Lifecycle / Tasks

Extract Transform Load.

D.E. (1) Data Ingestion → Getting the data / Loading {ETL}

D.A. (2) Data cleaning & processing → Clean data & processing → Majority (Exploratory Data Analysis (EDA)) Grunt work

D.S. (3) Building ML models → Training models

M.L.E (4) Deploying the models

BA / Consultants (5) Bridge the gap b/w tech & business.